

## Rationalising the denominator

Rationalise the denominator,

$$1) \frac{4 + \sqrt{3}}{3 + \sqrt{3}} = \frac{9 - \sqrt{3}}{6}$$

$$11) \frac{3 + \sqrt{6}}{2 + \sqrt{5}} = -6 + 3\sqrt{5} - 2\sqrt{6} + \sqrt{30}$$

$$2) \frac{2 + \sqrt{7}}{5 + \sqrt{6}} = \frac{10 - 2\sqrt{6} + 5\sqrt{7} - \sqrt{42}}{19}$$

$$12) \frac{3 + \sqrt{3}}{3 + \sqrt{2}} = \frac{9 - 3\sqrt{2} + 3\sqrt{3} - \sqrt{6}}{7}$$

$$3) \frac{4 + \sqrt{3}}{4 + \sqrt{5}} = \frac{16 - 4\sqrt{5} + 4\sqrt{3} - \sqrt{15}}{11}$$

$$13) \frac{3 + \sqrt{3}}{4 + \sqrt{6}} = \frac{12 - 3\sqrt{6} + 4\sqrt{3} - 3\sqrt{2}}{10}$$

$$4) \frac{4 + \sqrt{5}}{5 + \sqrt{3}} = \frac{20 - 4\sqrt{3} + 5\sqrt{5} - \sqrt{15}}{22}$$

$$14) \frac{4 + \sqrt{2}}{2 + \sqrt{7}} = \frac{-8 + 4\sqrt{7} - 2\sqrt{2} + \sqrt{14}}{3}$$

$$5) \frac{5 + \sqrt{5}}{5 + \sqrt{6}} = \frac{25 - 5\sqrt{6} + 5\sqrt{5} - \sqrt{30}}{19}$$

$$15) \frac{2 + \sqrt{3}}{4 + \sqrt{3}} = \frac{5 + 2\sqrt{3}}{13}$$

$$6) \frac{2 + \sqrt{2}}{3 + \sqrt{7}} = \frac{6 - 2\sqrt{7} + 3\sqrt{2} - \sqrt{14}}{2}$$

$$16) \frac{2 + \sqrt{7}}{3 + \sqrt{3}} = \frac{6 - 2\sqrt{3} + 3\sqrt{7} - \sqrt{21}}{6}$$

$$7) \frac{4 + \sqrt{5}}{4 + \sqrt{7}} = \frac{16 - 4\sqrt{7} + 4\sqrt{5} - \sqrt{35}}{9}$$

$$17) \frac{3 + \sqrt{2}}{5 + \sqrt{7}} = \frac{15 - 3\sqrt{7} + 5\sqrt{2} - \sqrt{14}}{18}$$

$$8) \frac{3 + \sqrt{3}}{3 + \sqrt{7}} = \frac{9 - 3\sqrt{7} + 3\sqrt{3} - \sqrt{21}}{2}$$

$$18) \frac{4 + \sqrt{7}}{3 + \sqrt{5}} = \frac{12 - 4\sqrt{5} + 3\sqrt{7} - \sqrt{35}}{4}$$

$$9) \frac{5 + \sqrt{7}}{5 + \sqrt{5}} = \frac{25 - 5\sqrt{5} + 5\sqrt{7} - \sqrt{35}}{20}$$

$$19) \frac{5 + \sqrt{7}}{2 + \sqrt{6}} = \frac{-10 + 5\sqrt{6} - 2\sqrt{7} + \sqrt{42}}{2}$$

$$10) \frac{3 + \sqrt{5}}{2 + \sqrt{6}} = \frac{-6 + 3\sqrt{6} - 2\sqrt{5} + \sqrt{30}}{2}$$

$$20) \frac{2 + \sqrt{2}}{5 + \sqrt{2}} = \frac{8 + 3\sqrt{2}}{23}$$